Ryotatsu Yanagimoto | Curriculum Vitae

Physics & Informatics Laboratories, NTT Research, Inc., Sunnyvale, CA 94085, USA
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Employment

NTT Research — **Sunnyvale, CA, USA**

Research Scientist — *Apr. 2024 – Present*

Postdoctoral fellow — *Apr. 2023 – Mar. 2024*

Cornell University — **Ithaca, NY, USA**

Visting scholar (PI: Prof. Peter L. McMahon) — *Apr. 2023 – Mar. 2026*

Education

Stanford University — **Stanford, CA, USA**

Ph.D. in Applied Physics (Research advisor: Prof. Hideo Mabuchi) — *Sep. 2017 – Apr. 2023*

- Dissertation title: "Quantum dynamics of broadband nonlinear photonics : from phenomenology to function"

The University of Tokyo — **Bunkyo-ku, Tokyo, Japan**

B.E. in Applied Physics (Research advisor: Prof. Hidetoshi Katori) — *Apr. 2013 – Mar. 2017*

- Dissertation title: "Characterization of collisional shifts in optical lattice clocks based on asymmetries in the Ramsey spectrum"

Publications and Preprints

* These authors contributed equally to the work

1. Sridhar Prabhu, Vladimir Kremenetski, Saeed A Khan, **Ryotatsu Yanagimoto**, Peter L McMahon, “Exponential advantage in quantum sensing of correlated parameters”, Quantum 10, 1963(2026).
2. Tatsuhiro Onodera, Martin M. Stein, Benjamin A. Ash, Mandar M. Sohoni, Melissa Bosch, **Ryotatsu Yanagimoto**, Marc Jankowski, Timothy P. McKenna, Tianyu Wang, Gennady Shvets, Maxim R. Shcherbakov, Logan G. Wright, Peter L. McMahon, “Arbitrary control over multimode wave propagation for machine learning”, Nature Physics 22, 164 (2026).

3. **Ryotatsu Yanagimoto**, Benjamin A. Ash, Mandar M. Sohoni, Martin M. Stein, Yiqi Zhao, Federico Presutti, Marc Jankowski, Logan G. Wright, Tatsuhiro Onodera, Peter L. McMahon, “Programmable on-chip nonlinear photonics”, Nature 649, 330 (2026).
4. **Ryotatsu Yanagimoto*,** Ouri Karni*, Edwin Ng, Marc Jankowski, Timothy P. Mckenna and Thibault Chervy, “Design and function of a vertical micro-cavity optical parametric oscillator”, Journal of Physics: Photonics 7, 045031(2025).
5. Valeria Cimini, Mandar M Sohoni, Federico Presutti, Benjamin K Malia, Shi-Yuan Ma, **Ryotatsu Yanagimoto**, Tianyu Wang, Tatsuhiro Onodera, Logan G Wright, Peter L McMahon, “Large-scale quantum reservoir computing using a Gaussian Boson Sampler”, arXiv:2505.13695.
6. Chris Gustin*, **Ryotatsu Yanagimoto***, Edwin Ng, Tatsuhiro Onodera and Hideo Mabuchi, “Effective field theories in broadband quantum optics: modeling phase modulation and two-photon loss from cascaded quadratic nonlinearities”, Quantum Science and Technology **10,** 025035 (2025).
7. **R. Yanagimoto*,** E. Ng*, M. Jankowski, R. Nehra, T. P. McKenna, T. Onodera, L. G. Wright, R. Hamerly, A. Marandi, M. M. Fejer, H. Mabuchi, “Mesoscopic ultrafast nonlinear optics—The emergence of multimode quantum non-Gaussian physics”, Optica **11**, 896 (2024).
8. Marc Jankowski*, **Ryotatsu Yanagimoto***, Edwin Ng, Ryan Hamerly, Timothy P McKenna, Hideo Mabuchi, MM Fejer, “Ultrafast second-order nonlinear photonics—from classical physics to non-Gaussian quantum dynamics: a tutorial”, Advances in Optics and Photonics **16,** 347 (2024).
9. E. Ng*, **R. Yanagimoto***, M. Jankowski, M. M. Fejer, H. Mabuchi, “Quantum noise dynamics in nonlinear pulse propagation”, arXiv:2307.05464.
10. **R. Yanagimoto***, R. Nehra*, E. Ng, A. Marandi, H. Mabuchi, “Engineering cubic quantum nondemolition Hamiltonian with mesoscopic optical parametric interactions”, arXiv:2305.03260.
11. **R. Yanagimoto***, R. Nehra*, R. Hamerly, E. Ng, A. Marandi, H. Mabuchi, “Quantum nondemolition measurements with optical parametric amplifiers for ultrafast universal quantum information processing”, PRX Quantum **4**, 010333 (2023).
12. **R. Yanagimoto**, E. Ng, M. Jankowski, H. Mabuchi, R. Hamerly, “Temporal trapping: a route to strong coupling and deterministic optical quantum computation”, Optica **9**, 1289 (2022).
13. **R. Yanagimoto***, E. Ng*, A. Yamamura, T. Onodera, L. G. Wright, M. Jankowski, M. M. Fejer, P. L. McMahon, H. Mabuchi, “Onset of non-Gaussian quantum physics in pulsed squeezing with mesoscopic fields”, Optica **9**, 379 (2022).
14. **R. Yanagimoto**, E. Ng, L. G. Wright, T. Onodera, H. Mabuchi, “Efficient simulation of ultrafast quantum nonlinear optics with matrix product states,” Optica **8**, 1306 (2021).

15. **R. Yanagimoto***, E. Ng*, T. Onodera, H. Mabuchi, “Towards an engineering framework for ultrafast quantum nonlinear optics,” Proc. SPIE 11684, Ultrafast Phenomena and Nanophotonics XXV, 116841D (2021).
16. **R. Yanagimoto***, E. Ng*, M. Jankowski, T. Onodera, M. M. Fejer, H. Mabuchi, “Broadband Parametric Downconversion as a Discrete-Continuum Fano Interaction,” arXiv:2009.01457.
17. **R. Yanagimoto*,** T. Onodera*, E. Ng, L. G. Wright, P. L. McMahon, H. Mabuchi, “Engineering a Kerr-based Deterministic Cubic Phase Gate via Gaussian Operations," Physical Review Letters **124**, 240503 (2020).
18. **R. Yanagimoto,** E. Ng, T. Onodera, H. Mabuchi, “Adiabatic Fock-state-generation scheme using Kerr nonlinearity," Physical Review A **100,** 033822 (2019).
19. **R. Yanagimoto,** P. L. McMahon, E. Ng, T. Onodera, H. Mabuchi, “Embedding entanglement generation within a measurement-feedback coherent Ising machine," arXiv:1906.04902 (2019).
20. N. Nemitz, A. A. Jørgensen, **R Yanagimoto**, F. Bregolin, H. Katori, "Modeling light shifts in optical lattice clocks," Physical Review A **99**, 033424 (2019). (Editors’ suggestion)
21. D. B. S. Soh, **R. Yanagimoto**, E. Chatterjee, H. Mabuchi, “Nonlinear optical response of a local surface plasmon coupled to a 2D material”, arXiv:1902.06943 (2019).
22. **R. Yanagimoto**, N. Nemitz, F. Bregolin, H. Katori, "Decomposed description of Ramsey spectra under atomic interactions," Physical Review A **98**, 012704 (2018).

Honors and Awards

Stanford Q-FARM Ph.D. Fellowship *2020 – 2022*

- Annual financial support of 50,000 USD for 2 years

Fellowship from Masason Foundation *2017 – 2022*

- Masason foundation is a public interest incorporated association founded by Masayoshi Son supporting "youth who will create the future."
- Financial support (entire tuition) for pursuing degree and research at Stanford University

Distinguished thesis award *Mar. 2017*

- Awarded by the Department of Applied Physics, the University of Tokyo for the undergraduate thesis research
- The award is given to distinguished thesis research of the year

Dean Award (Faculty of Engineering, The University of Tokyo) *Mar. 2017*

- The award is given to one graduating student with the best academic and research records in each department